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```
[2]: import pandas as pd
df = pd.read_csv('imdb250.csv')
print(df)
```

	Rank	Title	Year	Rating	Runtime
0	1	The Shawshank Redemption	1994	9.3	142
1	2	The Godfather	1972	9.2	175
2	3	The Dark Knight	2008	9.0	152
3	4	The Godfather Part II	1974	9.0	202
4	5	12 Angry Men	1957	9.0	96
..	...	...	...	...	...
245	246	Dersu Uzala	1975	8.0	142
246	247	Monsters Inc.	2001	8.0	92
247	248	Miracle in Cell No. 7	2019	8.0	132
248	249	Children of Heaven	1997	8.0	89
249	250	All About Eve	1950	8.0	138

[250 rows x 5 columns]

- Total no of movies

```
[5]: # 1. Total number of movies
print("1. Total number of movies:")
print(df.shape[0])
```

1. Total number of movies: ●●●

- Average IMDB rating

```
6]: # 2. Average IMDb rating
print("\n2. Average IMDb rating:")
print(df['Rating'].mean())
```

2. Average IMDb rating:
8.253200000000001

- Movie with highest rating

```
[7]: # 3. Movie with the highest rating
print("\n3. Movie(s) with the highest rating:")
print(df[df['Rating'] == df['Rating'].max()])
```

```
3. Movie(s) with the highest rating:
Rank      Title      Year  Rating  Runtime
0         1  The Shawshank Redemption  1994      9.3      142
```

- Oldest movie

```
Oldest movie
("\n4. Oldest movie:")
(df[df['Year'] == df['Year'].min()])
```

```
Oldest movie:
Rank      Title      Year  Rating  Runtime
100      The Kid  1921      8.3      68
```

- Newest movie

```
[10]: # 5. Newest movie
print("\n5. Newest movie:")
print(df[df['Year'] == df['Year'].max()])
```

```
5. Newest movie:
Rank      Title      Year  Rating  Runtime
117      118  Hamilton  2020      8.2      160
```

- Missing values in each column

```
6. Missing values in each column
print("\n6. Missing values in each column:")
print(df.isnull().sum())
```

```
Missing values in each column:
Rank      0
Title     0
Year      0
Rating    0
Runtime   0
dtype: int64
```

- Movies with rating higher than 9.0

```
[16]: # 7. Movies with rating greater than 9.0
print("\n7. Movies with rating greater than 9.0:")
print(df[df['Rating'] > 9.0])
```

```
7. Movies with rating greater than 9.0:
Rank      Title      Year  Rating  Runtime
0         1  The Shawshank Redemption  1994      9.3      142
1         2      The Godfather  1972      9.2      175
```

- Movies released before 1970

```
]: # 8. Movies released before 1970
print("\n8. Movies released before 1970:")
print(df[df['Year'] < 1970])
```

8. Movies released before 1970:

	Rank	Title	Year	Rating	Runtime
4	5	12 Angry Men	1957	9.0	96
9	10	The Good the Bad and the Ugly	1966	8.8	178
18	19	Seven Samurai	1954	8.7	207
23	24	It's a Wonderful Life	1946	8.6	130
39	40	Casablanca	1942	8.5	102
..	...	...	...	...	...
225	226	Persona	1966	8.0	85
234	235	White Heat	1949	8.0	114
240	241	The Red Shoes	1948	8.0	135
242	243	The Hustler	1961	8.0	134
249	250	All About Eve	1950	8.0	138

[66 rows x 5 columns]

- Movies with longest runtime

```
[29]: # 9. Movie(s) with the longest runtime
print("\n9. Movie(s) with the longest runtime:")
print(df[df['Runtime'] == df['Runtime'].max()])
```

9. Movie(s) with the longest runtime:

	Rank	Title	Year	Rating	Runtime
233	234	Fanny and Alexander	1982	8.0	312

- Standard deviation of movie ratings

```
# 13. Standard deviation of movie ratings
print("\n13. Standard deviation of ratings:")
print(df['Rating'].std())
```

13. Standard deviation of ratings:
0.255874278868247

- Group by decade and find average rating per decade

```
[32]: # 11. Group by decade and find average rating per decade
df['Decade'] = (df['Year'] // 10) * 10
print("\n11. Average rating per decade:")
print(df.groupby('Decade')['Rating'].mean())
```

11. Average rating per decade:

Decade	
1920	8.150000
1930	8.250000
1940	8.192308
1950	8.233333
1960	8.235294
1970	8.300000
1980	8.213333
1990	8.335556
2000	8.259574
2010	8.217073
2020	8.200000

Name: Rating, dtype: float64

- Median year of movie releases

```
[38]: # 17. Median year of movie releases
print("\n17. Median year of movie releases:")
print(df['Year'].median())
```

17. Median year of movie releases:
1993.0

- Number of movies released each year

```
[33]: # 12. Number of movies released each year
print("\n12. Number of movies released each year:")
print(df['Year'].value_counts().sort_index())
```

12. Number of movies released each year:

Year	
1921	1
1924	1
1925	1
1926	1
1927	1
..	
2016	3
2017	3
2018	2
2019	6
2020	1

Name: count, Length: 84, dtype: int64

- Correlation between year and rating

```
# 16. Correlation between year and rating
print("\n16. Correlation between Year and Rating:")
print(df['Year'].corr(df['Rating']))
```

16. Correlation between Year and Rating:
0.043096773335916806

- Movies with runtime greater than 150 minutes

```
[36]: # 15. Movies with runtime greater than 150 minutes
print("\n15. Movies with runtime greater than 150 minutes:")
print(df[df['Runtime'] > 150])
```

15. Movies with runtime greater than 150 minutes:

Rank	Title	Year	Rating	\
1	2	The Godfather	1972	9.2
2	3	The Dark Knight	2008	9.0
3	4	The Godfather Part II	1974	9.0
5	6	Schindler's List	1993	9.0
6	7	The Lord of the Rings: The Return of the King	2003	8.9
7	8	Pulp Fiction	1994	8.9
8	9	The Lord of the Rings: The Fellowship of the Ring	2001	8.8
9	10	The Good the Bad and the Ugly	1966	8.8
13	14	The Lord of the Rings: The Two Towers	2002	8.8
18	19	Seven Samurai	1954	8.7
25	26	Saving Private Ryan	1998	8.6
27	28	The Green Mile	1999	8.6
29	30	Interstellar	2014	8.6
32	33	Gladiator	2000	8.5
34	35	The Departed	2006	8.5
43	44	Cinema Paradiso	1988	8.5
44	45	Once Upon a Time in the West	1968	8.5
50	51	Django Unchained	2012	8.4
60	61	The Dark Knight Rises	2012	8.4
69	70	Braveheart	1995	8.4
71	72	3 Idiots	2009	8.4
73	74	Like Stars on Earth	2007	8.4
75	76	Amadeus	1984	8.3
76	77	Inglourious Basterds	2009	8.3
86	87	Lawrence of Arabia	1962	8.3
94	95	Scarface	1983	8.3
104	105	Metropolis	1927	8.3
114	115	The Great Escape	1963	8.2
116	117	Heat	1995	8.2
117	118	Hamilton	2020	8.2

	Runtime	Decade
1	175	1970
2	152	2000
3	202	1970
5	195	1990
6	201	2000
7	154	1990
8	178	2000
9	178	1960
13	179	2000
18	207	1950
25	169	1990
27	189	1990
29	169	2010
32	155	2000
34	151	2000
43	155	1980
44	165	1960
50	165	2010
60	164	2010
69	178	1990
71	170	2000
73	162	2000
75	160	1980
76	153	2000
86	228	1960
94	170	1980
104	153	1920
114	172	1960
116	170	1990
117	160	2020
118	156	2000
120	162	1980
121	180	2010
123	178	1990
126	158	2000

- Year of movie releases

```

year of movie releases
Median year of movie releases:")
ar'].median())

```

ear of movie releases:

- Top 5 Years with Highest Average IMDb Rating

```

[46]: # 18. Top 5 Years with Highest Average IMDb Rating
print("\n18. Top 5 Years with Highest Average IMDb Rating:")
print(df.groupby('Year')['Rating'].mean().sort_values(ascending=False).head(5))

```

```

18. Top 5 Years with Highest Average IMDb Rating:
Year
1972    9.2
1994    8.8
1990    8.7
1946    8.6
1977    8.6
Name: Rating, dtype: float64

```

- Average runtime

```

[42]: # 19. Average runtime
print("\n19. Average runtime of movies (in minutes):")
print(df['Runtime'].mean())

```

```

19. Average runtime of movies (in minutes):
129.108

```

- Movies with shortest runtime

```
[31]: # 10. Movie(s) with the shortest runtime
print("\n10. Movie(s) with the shortest runtime:")
print(df[df['Runtime'] == df['Runtime'].min()])
```

```
10. Movie(s) with the shortest runtime:
   Rank  Title  Year  Rating  Runtime
182  183  Sherlock Jr.  1924    8.1    45
```

- Pivot table showing average runtime for each rating

```
[40]: # 20. Pivot table showing average runtime for each rating
print("\n20. Pivot table showing average runtime for each rating:")
print(df.pivot_table(values='Runtime', index='Rating', aggfunc='mean').round(1))
```

```
20. Pivot table showing average runtime for each rating:
```

Rating	Runtime
8.0	125.4
8.1	121.9
8.2	136.8
8.3	124.4
8.4	131.6
8.5	119.5
8.6	138.7
8.7	149.2
8.8	160.7
8.9	177.5
9.0	161.2
9.2	175.0
9.3	142.0